TEXORA · SESSIONAL NOTES · STUDY SHEET

Production of Suits, Jacket, Jogging Suit, Dresses, Children Wear and Night Wear

Producing six garment categories in the lab end to end — pattern, cutting, assembly sequence, pressing and inspection — and seeing how construction complexity changes from a tailored suit to a simple night dress.

AM-4202 · ADVANCED APPAREL MANUFACTURING-II SESSIONAL · STUDY 3 · LEVEL 4 TERM 2

1. Aim and place of the study

Aim: to plan and carry out the complete production of representative garments from six categories — a suit (jacket with trouser), a separate jacket, a jogging suit, dresses, children's wear and night wear — recording for each the pattern pieces, the assembly sequence, the machines used and the quality checkpoints.

The six categories were chosen because they span the industry's construction range: the **suit** is the most structured garment in ordinary production (canvas, padding, shaped pressing), the **jogging suit and night wear** are among the simplest (knit, elasticated, few components), and **dresses and children's wear** add their own rules — children's wear above all carries safety requirements that adult garments do not.

2. Materials and reference measurements

Materials: suiting with fusible interlining and chest canvas for the jacket; matching trouser fabric; knit fleece or interlock for the jogging suit; woven and knit dress fabrics; printed cotton for children's wear; soft knit or cotton for night wear; standard trims (zips, buttons, elastic, drawcord, labels).

Reference charts used across Texora apparel studies (garment values, cm). Men's basic shirt, size M: chest 107, length 73, shoulder 46, sleeve 61.5, collar 40 (grades 5/2/2/1.5/2). Men's basic trouser, size M: waist 82, hip 102, outseam (incl. waistband) 104, inseam 78, thigh 62, knee 46, hem width 40, front rise 26; seam allowances 1 cm general, 1.5 cm at side seams and waistband, 2.5 cm at hems. Specimens in this study are made to these charts where the garment matches them; the other categories use the lab's standard blocks for their type, recorded in the report.

3. The common production route

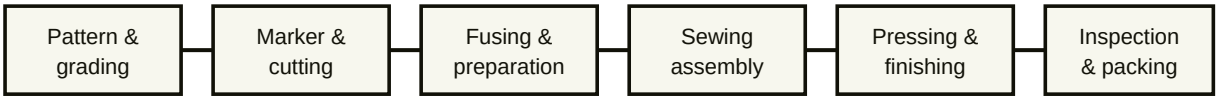


Figure 1 — The route every category follows. What changes between categories is the weight of each stage: fusing and pressing dominate the suit; assembly speed dominates the knit garments.

1. **Pattern and grading.** Draft or adapt the block for the category, add seam allowances per the study rule (1 cm general, 1.5 cm at adjustment seams, 2.5 cm hems), grade to the size range, and mark grain lines, notches and drill holes.
2. **Marker and cutting.** Lay the pieces in a marker (efficiency measured in Study 4); cut with straight-knife or band-knife for tailored parts, die or round knife for knits; number and bundle the parts.
3. **Fusing and preparation.** Apply fusible interlining to jacket fronts, collars, cuffs and waistbands at the press settings the interlining maker specifies; prepare canvases, linings, pockets and trims.
4. **Sewing assembly.** Build the garment in sub-assemblies (fronts, backs, sleeves, collar, lining) and join them in the category sequence of §4; SNLS for seams, overlock for edge neatening and knit joining, buttonhole and button-attach machines at the end (machines as studied in Study 1).
5. **Pressing and finishing.** Press in stages — under-pressing of parts during assembly, final top-pressing of the finished garment — then trim threads, attach tickets and labels.
6. **Inspection and packing.** Measure against the chart, inspect visually by defect class (critical / major / minor, as in TE-3107), fold and pack.

4. Category notes: what is different in each

CATEGORY	CONSTRUCTION ESSENTIALS RECORDED IN THIS STUDY
Suit (jacket + trouser)	The jacket front is built up with fusible and a chest canvas so the chest keeps a shaped, rolled form; lapels are pad-stitched or fused to roll, not lie flat; sleeves are set with fullness eased into the armhole; pockets (welt, flap, inside) are made before the fronts are closed; lining is assembled separately and joined so the garment hangs from the outer shell; trousers follow the TE-3108 sequence — fly, pockets, waistband, crease pressing. Final pressing on shaped forms is part of the construction, not a cosmetic extra.
Jacket (separate)	Shares the suit jacket's structure with a casual specification: softer interlining or none, patch pockets in place of welt, unlined or half-lined body — the study records how

	each softening step changes both appearance and work content.
Jogging suit	Knit two-piece (top and bottom) built almost entirely on overlock and flatlock: ribbed cuffs, hem band and collar or hood; kangaroo pocket on the top; elastic-and-drawcord waist on the bottom; no fusing, no lining, minimal pressing. The contrast with the suit — hours of structure against minutes of assembly — is the study's main lesson in work content.
Dresses	One-piece woven or knit garments: bodice and skirt joined at the waist (or cut in one), zip or button closure, darts and gathers for shaping, hem finishes from narrow rolled to faced. Fit at bust–waist–hip is the control point; the study measures all three against the block.
Children's wear	Construction follows the adult pattern logic with **safety rules added**: no drawstrings or cords at hood and neck in young children's garments (entanglement hazard), small parts (buttons, snaps) must pass pull-strength discipline and be sewn to standard, no sharp trims; prints and dyes chosen for skin contact. A found needle fragment or a loose button is a **critical** defect here, as in the TE-3107 classification — zero tolerance.
Night wear	Pyjama sets and night dresses in soft cotton or knit: comfort construction — flat or outside-lying seams against the skin, elasticated waists, no scratchy labels at the neck (printed labels used), generous ease. Where the market requires it, flammability behaviour of the fabric is part of the specification, and trims are kept minimal.

5. Worked example: fabric requirement for the trouser component

The marker for the basic trouser (size M, chart above) at 150 cm fabric width gives a marker length of **1.12 m per garment** for the lab's size mix. For the study order of **2,400 trousers** (the same lot size used in the TE-3107 AQL example and in Study 6):

$$\text{fabric} = 2{,}400 \times 1.12\ \text{m} = 2{,}688\ \text{m};\ \text{adding 3\% cutting-room allowance} = 2{,}688 \times 1.03 = 2{,}768.6\ \text{m}$$

The purchase quantity is therefore ordered as **2,769 m** of the trouser fabric (plus the pocketing, waistband and interlining quantities computed the same way from their own markers). The same ×(1 + allowance) structure is used for every category: marker length fixes the net consumption, and the allowance — never a guess hidden inside the marker — is stated on its own line so it can be audited.

Precautions

- Check fusing temperature, pressure and time on a test piece before fusing garment fronts: under-fusing peels in wear, over-fusing strike-through ruins the face fabric.

- Keep tailored parts on grain at every stage; a jacket front cut off-grain cannot be pressed back into shape.
- Children's wear specimens: apply the safety rules of §4 as hard checks, not preferences — cords at the neck and untested small parts stop the specimen.
- Needle discipline in all categories: broken-needle procedure and metal-free verification of finished children's and night wear before packing.

Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why does a suit jacket carry canvas and interlining while a jogging top carries none?
2. State two safety rules that apply to children's wear but not to adult garments, and the defect class of a breach.
3. A marker gives 0.86 m per jogging bottom. For 5,000 pieces with a 3% allowance, how much fabric is ordered?
4. Which stages of the common route grow, and which shrink, as construction moves from the suit to night wear?

Answer key

1. The jacket is a shaped, structured garment: canvas and interlining create and hold the three-dimensional chest and lapel roll against gravity and wear. The jogging top is a soft knit garment whose shape comes from the pattern and the rib trims; structure would destroy the comfort it is designed for.
2. No drawstrings/cords at hood and neck (entanglement), and small parts fixed to pull-strength discipline with no sharp trims. A breach — for example a loose button or a needle fragment — is a critical defect: zero tolerance, the garment cannot ship.
3. Fabric = 5,000 × 0.86 = 4,300 m; with allowance 4,300 × 1.03 = 4,429 m ordered.
4. Fusing/preparation and pressing grow sharply for the suit (canvas building, under-pressing, shaped top-pressing) and almost vanish for night wear; assembly stays the core stage in both, but in soft garments it is fast overlock work, while the suit's assembly is slow, multi-machine and interrupted by pressing.
